

Characterisation and Quantification of Data Centre Flexibility for Power System Support

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Highlights

- Linked annual optimisation preserves physical and workload continuity.
- Integrated operation cuts annual electricity cost by 5.049%.
- Flexible workload has greater value than UPS or TES capacity.
- Recovery-certified flexibility reaches 9.75 hours on a typical day.
- Component decomposition identifies how each grid response is delivered.

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Abstract—The rapid growth of data centres creates both a power-system challenge and an opportunity for demand-side flexibility. This paper develops a linked rolling-horizon mixed-integer optimisation of IT workload scheduling, uninterruptible power supply (UPS) dispatch, cooling dynamics and thermal energy storage (TES) for a 1 MW data centre. Physical states and unfinished workload cohorts are carried exactly between local-day horizons across a full year, including daylight-saving transitions. Against a workload-at-arrival, storage-disabled benchmark with cost-optimised cooling, coordinated operation reduces 2025 electricity cost from GBP 550,503.59 to GBP 522,707.23, a saving of 5.049%. Full-year endpoints rank flexible workload as the most influential tested resource, followed by UPS and TES energy capacity. An objectively selected representative day is then subjected to 288 grid-deviation requests. Each request begins from the stored annual state, follows the annual pre-event CPU trajectory and must recover storage and thermal states after the event. The resulting duration envelope is strongly time- and direction-dependent: increased import is sustained for up to 9.75 hours at +25 kW, whereas reduced import is sustained for up to 5.25 hours at -100 and -150 kW. An additive component decomposition identifies the contributions of IT load, CRAC cooling, TES charging and UPS dispatch. All continuity, workload, exclusivity, settlement and flexibility-response audits pass. Of 288 cells, 283 have exact duration boundaries and five retain explicitly marked certified lower bounds, establishing a reproducible link between annual economic operation and recovery-certified grid flexibility.

Index Terms—Data Centre, Power system flexibility, Demand side flexibility, Load Shifting, Thermal Inertia

I. INTRODUCTION

GLOBAL digitalisation, propelled by artificial intelligence, is driving the rapid expansion of data centres, whose electricity demand is both substantial and spatially concentrated. This growth coincides with two other major shifts in the energy landscape: the increasing share of variable renewable generation needed for decarbonisation, and the widespread adoption of new demand-side technologies such as electric vehicles and heat pumps [1], [2]. The combined effect places significant strain on modern power systems

by intensifying peak loads, steepening net load ramps, and aggravating local network constraints. Consequently, there is an escalating need for power system flexibility to maintain the crucial balance between electricity supply and demand.

According to the IEA, global data centre electricity consumption was around 415 TWh in 2024, with projections suggesting this will exceed 945 TWh by 2030, raising their share of global electricity demand from approximately 1.5% to 3% [3]. This expansion is not uniform, with over 80% of the increase expected to originate from the United States and China [3]. Moreover, geographical clustering intensifies the impact on local grids. For instance, data centre consumption already accounts for over 20% of national demand in Ireland, and 25% in Virginia, USA [3]. In Great Britain, annual electricity consumption by data centres is forecast by National Grid ESO to rise from 7.6 TWh in 2024 to between 30–71 TWh by 2050 [4].

To manage the volatility introduced by these trends, power systems require a substantial increase in flexibility services. Traditionally provided by dispatchable fossil fuel generators, this solution is becoming obsolete under decarbonisation mandates. The JRC [5] projects that the European Union's flexibility requirement will rise to 24% of total electricity demand in TWh by 2030 and to 30% by 2050. Similarly, Great Britain's Clean Flexibility Roadmap targets an expansion of flexibility capacity from 25.2 GW to between 54–66 GW by 2030 [6]. Demand-side flexibility, where consumers actively modulate their electricity use to support the grid, presents a key solution.

While their growth contributes to the challenge, data centres are equipped to provide flexibility. Their shiftable IT workloads, uninterruptible power supply (UPS) systems with battery storage, and inherent thermal inertia as well as cold storage, can be leveraged to modulate power consumption without disrupting core services. Each of these internal assets possesses a distinct flexibility profile in terms of magnitude, duration, ramp rate, and recovery time [7]. For example, UPS

batteries can offer a near instantaneous response, whereas thermal systems are characterised by lower ramp rates, providing a more gradual adjustment [7], [8]. Data centres therefore present a dichotomy, they are both a growing strain on the power system and a potentially crucial asset for grid flexibility.

In the context of power systems, flexibility is defined as the capability of maintaining the balance between generation and demand across all time horizons, by adjusting either energy production or consumption in response to sudden variations [1], [2], [4], [9]. Upward flexibility is required when demand exceeds generation and involves a DC reducing its power consumption from the grid. Conversely, downward flexibility is needed when generation surpasses demand, and is provided by a DC increasing its electricity consumption to absorb the surplus energy [1], [10]. In this study, a data centre's flexibility is quantified by its capacity for power consumption deviation relative to a baseline, and the maximum duration for which this deviation can be sustained.

This paper addresses these challenges with a linked annual optimisation of IT workload, UPS dispatch and cooling operation against signed electricity prices. Its principal contribution is an auditable rolling-horizon formulation that passes physical states and unfinished workload between local days instead of resetting the facility at midnight. The study quantifies annual cost value, tests full-year sensitivity endpoints for the three internal flexibility resources, and defines a recovery-constrained duration assessment to be applied to an objectively selected representative day.

The main contributions are therefore: (1) a tractable integrated data-centre model that preserves the relevant coupling between IT execution, UPS operation, cooling power and TES state; (2) a complete 2025 rolling chain with explicit state, workload, solver and settlement audits; and (3) annual sensitivity evidence that separates full-year effect sizes from representative-week response shapes.

The remainder of this paper reviews the literature, defines the linked annual method and benchmark, presents the completed annual results and sensitivities, and identifies the deferred representative-day flexibility assessment.

A. Data Centre Architecture

DCs are integrated facilities housing IT equipment, such as servers and network devices, which are supported by critical power and cooling infrastructure. As shown in Figure 1, electrical continuity is maintained by uninterruptible power supplies (UPS) with associated battery systems, while cooling systems, including chillers, Thermal Energy Storage (TES) tanks and Computer Room Air Conditioning (CRAC) units, manage the substantial heat generated by the servers. Thermal efficiency is often enhanced through optimised airflow strategies like the hot-aisle/cold-aisle arrangement.

Each of these core subsystems presents an opportunity for providing demand-side flexibility to the power grid. The strategic scheduling of IT workloads, the dispatch of UPS battery storage, and the leveraging of the TES tank and the facility's inherent thermal inertia allow a DC to modulate its power consumption. The DC assets are highly interdependent,

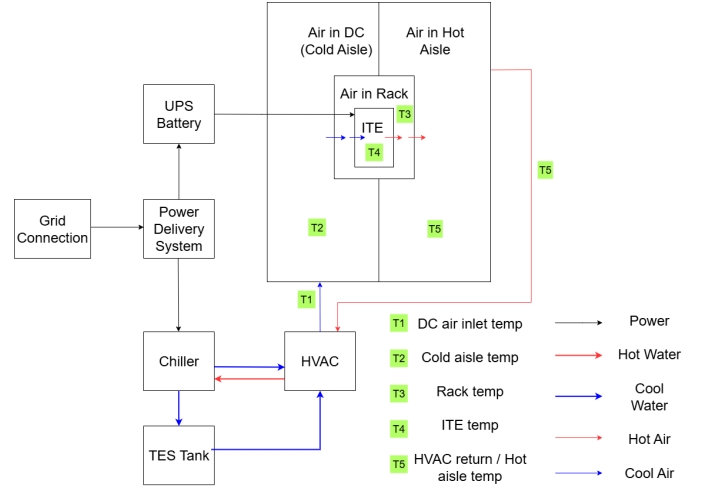


Fig. 1. Data Centre Layout

requiring an integrated approach that considers all of them simultaneously. This is because a change in one subsystem changes the feasible operating range of the others: deferring IT workload reduces both server power and subsequent heat generation, cooling actions are constrained by thermal states and the TES charge level, and UPS discharge can support IT execution while altering the grid power draw. A component-by-component assessment can therefore overestimate flexibility if it ignores recovery requirements, or underestimate it if it misses compensating actions across assets. Consequently, DCs can transition from being passive electrical loads to active assets that enhance the resilience and flexibility of the power system.

B. IT Workload Background

The fundamental role of a DC is to execute IT workloads, which encompass all computational and data processing tasks. These workloads can be broadly classified based on their time sensitivity, as defined by service-level agreements (SLAs) [11], [12]. Inflexible workloads, also known as interactive workloads, are latency-critical and require immediate execution to ensure quality of service for user-facing applications like real-time streaming or transaction processing.

In contrast, flexible IT workloads, often referred to as batch workloads, can tolerate execution delays within predefined time windows without impacting service quality. These typically non-user-facing tasks, such as periodic data analytics, backups, or machine learning model training, can be deferred or rescheduled. This ability to shift the timing of their execution is the primary source of IT-based flexibility, allowing the DC to adjust its computational power draw in response to external signals like electricity prices or grid operational needs [12].

II. LITERATURE REVIEW

Data centre flexibility has attracted growing attention as an enabler of demand-side participation in power systems, with a growing body of literature modelling flexibility at the asset level, predominantly focusing on IT workloads and, to a lesser

extent, UPS batteries, cooling systems and other subsystems. A smaller body of work has attempted integrated models that capture the joint flexibility of multiple DC assets. This review therefore first presents the flexibility modelling approaches found in the literature for each DC asset, and then discusses studies that move towards an integrated data centre flexibility assessment.

IT workload is the most fundamental component of a DC and is the focus for many flexibility studies. One study develops a machine learning method to predict DC energy consumption and server temperature [13]. These data are then used to optimise the scheduling of IT workloads to minimise fuel costs and power utilisation at times of high grid carbon intensity. On average, the implementation achieved a 6.5% load reduction for a winter day during periods of high CO₂ intensity. Radovanović et al. [14], performed predictive modelling of grid carbon intensity, trained day-ahead demand prediction models and used them to optimise IT workload for minimum carbon emissions. Cao et al. [15] propose and validate a data-driven methodology using real-world Alibaba trace data to assess the power flexibility potential of Internet Data Centres (IDCs) by rescheduling periodic IT workloads. Their findings quantify significant flexibility, particularly in early morning hours, revealing potential upward/downward load shifts of 400kW and 325kW, respectively—corresponding to approximately 50% and 40.6% of the facility’s peak demand—by optimising job schedules based on power system needs and renewable energy availability. In this study, IT workloads are defined as either flexible or inflexible, with flexibility here referring to flexibility in time.

UPS batteries are currently providing power flexibility to the grid in numerous DCs around the world and have been shown to be successful [16]–[18]. Lithium-ion batteries, which are increasingly the dominant form used for UPS, are capable of rapidly providing or consuming power when needed [8]. These characteristics make them well suited to frequency regulation, a service many power providers require. Results from [8] found UPS batteries could be used within their normal operating range to provide the required services to the grid and in doing so generate a reasonable revenue stream. Furthermore, the battery lifetime would not be affected, and the approach does not require additional cost.

Cooling makes up a large proportion, as high as 40%, of DC energy consumption [19]. The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) recommends maintaining an indoor temperature of 18°C to 27°C to ensure optimal DC performance [20]. Exploiting the thermal inertia provided by this temperature range has been shown to enable power consumption flexibility in DCs [21] [22]. Thermal Energy Storage (TES) systems can enhance cooling flexibility by decoupling the timing of heat removal from the cooling generation process. Du et al. [23] propose and evaluate a framework to enhance the energy flexibility of district heating (DH) systems integrated with DC waste heat recovery. The study utilises dual short-term TES tanks to enable simultaneous peak shaving (demand-driven) and load shifting (price-driven) within the hybrid system. Simulation results demonstrated that this dual TES approach significantly

improved flexibility, achieving up to a 10% peak demand reduction and a 2.1% load shift, leading to a 3.2% operational cost saving in the case study. A ‘synergistic control strategy for data centre frequency regulation which uses both IT and cooling systems’ has also been proposed [24]. Results found that a revenue saving of 4% could be made by implementing the proposed strategy. Once depleted, the thermal inertia of the DC atmosphere / TES must be recovered to baseline levels. The recovery time taken to do so will affect how the flexibility of the asset can be utilised.

While individual assets offer flexibility, the flexibility characteristics change when these assets are integrated into one system. Maximising the overall potential requires coordinating multiple resources and evaluating their diverse characteristics, interactions, and constraints. One study proposes a co-optimisation of the IT workload and cooling system, recognising the strong operational link between them [24]. The work by S Xiang et al. [25] identifies that data centre optimisation models often ignore IT equipment constraints like start-stop conditions and ramp rates, which can cause excessive wear. The authors propose a model integrating these constraints with facility thermal dynamics, solved using a Deep Deterministic Policy Gradient (DDPG) algorithm. This approach was then validated for practicality and efficiency using a large-scale simulation of a 100,000-device data centre. H. Xu et al. [26] propose an optimisation which integrates IT servers, cooling infrastructure, energy storage, generators and renewable energy, and aims to maximise power demand flexibility. The objective function is the cumulative change in energy taken from the grid before and after the optimisation was applied. Other studies use a similar approach, but instead minimise the difference between DC power consumption and the desired power consumption of the power supply operator [27].

While individual DC assets, including IT workloads, UPS batteries and cooling systems, have each been studied for their flexibility potential, the interplay between these assets and its impact on the overall flexibility of the DC remains largely unexplored. Most existing studies adopt a DC-centric, cost-minimisation perspective and do not quantify the magnitude and duration of grid-facing flexibility that can be delivered once coupled asset constraints and post-event recovery are accounted for. As DCs scale towards the gigawatt level, this power-system-centric perspective becomes increasingly important, because system operators require duration-certified flexibility offers and a clear understanding of which assets enable them; this perspective remains largely absent from the literature.

Addressing these gaps, this study makes three contributions. First, it establishes an audited full-year operating baseline in which physical states and deadline-constrained work are linked between daily horizons. Second, it quantifies how workload flexibility, UPS capacity and TES capacity affect the annual economic result using completed full-year endpoints. Third, it maps the maximum recovery-certified duration of grid-power deviations on an objectively selected day and decomposes each response into its IT, cooling, TES and UPS contributions. The annual and event studies use the same stored trajectory, so

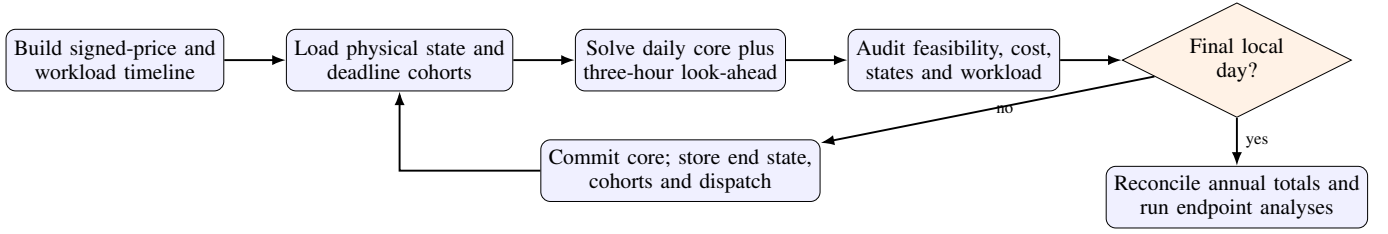


Fig. 2. Restartable linked rolling-horizon workflow. Stored checkpoints are the sole source of opening states for subsequent local days and for the representative-day flexibility assessment.

the reported service envelope is conditional on an internally consistent cost-optimal operating state rather than an independently reset day.

III. METHODOLOGY

A. Modelling scope and annual workflow

The data centre is represented as an aggregate 1 MW facility because the quantity of interest is grid-facing flexibility rather than server-level placement. The retained states are those that constrain service delivery: deadline-indexed computational work, UPS and TES energy, four facility temperatures and total grid import. This abstraction preserves the coupling between computation, electrical storage and cooling while remaining tractable for a year of repeated mixed-integer linear programmes (MILPs).

The model uses 15-minute intervals. Each local calendar day is the committed core of one solve and is followed by a 12-interval (three-hour) look-ahead. Consequently, daylight-saving days contain 92 or 100 core intervals and normal days contain 96. Only the core contributes to reported calendar-year cost; the look-ahead exposes every flexible-work deadline and establishes a feasible continuation. At the end of day d , the vector

$$\mathbf{x}_d^{\text{end}} = (E^{\text{UPS}}, E^{\text{TES}}, T_{\text{IT}}, T_{\text{R}}, T_{\text{CA}}, T_{\text{HA}}) \quad (1)$$

is imposed as the initial state of day $d + 1$. Outstanding workload is passed as cohorts retaining its original arrival time and absolute deadline.

Figure 2 summarises the implementation. Every committed day is written with a checkpoint and audit record, allowing an interrupted run to resume without reinitialising the system. An absolute UTC timeline prevents duplicate or missing timestamps, while local dates define settlement cores.

B. Input data and workload construction

The price signal is the actual signed 2025 Intermittent Market Reference Price (IMRP), expanded from hourly observations to quarter-hour intervals. Negative prices are retained. The workload is a stylised local-clock profile synthesised from the utilisation and flexible-share ranges reported for Google and Alibaba traces and related aggregate workload studies [12], [28]–[32]. Reported tabular values were transcribed and graphical profiles digitised; the hourly minima, maxima and means were compared, after which a representative profile inside the observed envelope was selected while preserving the source balance between inflexible and flexible demand. The

resulting trace is therefore a literature-synthesised modelling assumption, not a measurement of a particular facility. It repeats in local clock time throughout the year.

At each arrival interval, flexible demand is divided equally among four delay classes with maximum deferrals of 0.5, 1, 2 and 3 hours. If a_c is the CPU-hours arriving in cohort c , $u_{c,t}$ its scheduled utilisation and $\mathcal{W}_c$ its feasible execution window, then

$$\sum_{t \in \mathcal{W}_c} u_{c,t} \Delta t = a_c, \quad (2)$$

$$u_t^{\text{inflex}} + \sum_c u_{c,t} \leq 1, \quad u_{c,t} \geq 0. \quad (3)$$

Any balance whose deadline lies beyond the committed day is carried forward. The complete three-hour look-ahead is as long as the maximum permitted delay, so every cohort created in the core has a visible deadline.

Aggregate IT electrical power follows

$$P_t^{\text{IT}} = P_{\text{idle}} + (P_{\text{max}} - P_{\text{idle}}) u_t^{1.32}, \quad (4)$$

where $P_{\text{idle}} = 166.7$ kW and $P_{\text{max}} = 1000$ kW. A four-segment non-uniform DLOG piecewise-linear representation is used in the MILP. Its maximum error over $0 \leq u \leq 1$ is 6.241 kW (0.624% of rated IT power).

C. UPS, TES and thermal dynamics

The central UPS has 600 kWh capacity, a 50% minimum state of charge, charge and discharge efficiencies of 0.82 and 0.92, and a 270 kW charging limit. Its recursion is

$$E_{t+1}^{\text{UPS}} = E_t^{\text{UPS}} + \eta_{\text{ch}} P_{\text{ch},t}^{\text{UPS}} \Delta t - \frac{P_{\text{dis},t}^{\text{UPS}}}{\eta_{\text{dis}}} \Delta t. \quad (5)$$

A binary mode variable prevents simultaneous charge and discharge. UPS discharge supplies IT behind the meter, so $P_t^{\text{grid,IT}} = P_t^{\text{IT}} - P_{\text{dis},t}^{\text{UPS}}$.

The 1000 kWh-th TES has 300 kW-th charge and discharge limits and 0.9 efficiencies in both directions:

$$E_{t+1}^{\text{TES}} = E_t^{\text{TES}} + \left(\eta_{\text{ch}}^{\text{TES}} Q_{\text{ch},t}^{\text{TES}} - \frac{Q_{\text{dis},t}^{\text{TES}}}{\eta_{\text{dis}}^{\text{TES}}} \right) \Delta t. \quad (6)$$

Charging and discharging are mutually exclusive. The chiller coefficient of performance is 5 and its total electrical input is limited to 400 kW:

$$Q_t^{\text{cool}} = \text{COP} P_t^{\text{ch,CRAC}} + Q_{\text{dis},t}^{\text{TES}}, \quad (7)$$

$$Q_{\text{ch},t}^{\text{TES}} = \text{COP} P_t^{\text{ch,TES}}. \quad (8)$$

The four-node IT, rack, cold-aisle and hot-aisle model uses the heat capacities, air flow and conductances documented in the facility model. Backward Euler is used because the configured air-flow time constants make a 15-minute explicit update unstable. For example, the IT and cold-aisle balances are

$$T_{t+1}^{\text{IT}} = T_t^{\text{IT}} + \frac{\Delta t_s}{C_{\text{IT}}} [1000P_t^{\text{IT}} - G_{\text{conv}}(T_{t+1}^{\text{IT}} - T_{t+1}^{\text{R}})], \quad (9)$$

$$T_{t+1}^{\text{CA}} = T_t^{\text{CA}} + \frac{\Delta t_s}{C_{\text{CA}}} [\dot{m}\kappa c_p(T_t^{\text{in}} - T_{t+1}^{\text{CA}}) - G_{\text{cold}}(T_{t+1}^{\text{CA}} - T^{\text{out}})]. \quad (10)$$

The rack and hot-aisle balances use the same implicit end states. Cold-aisle temperature is constrained to 18–23°C, outdoor temperature is fixed at 22°C, and

$$Q_t^{\text{cool}} \geq 1000P_t^{\text{IT}} \quad (11)$$

prevents unvalidated accumulation of IT heat from being treated as a free flexibility resource.

D. Grid balance, objective and annual cases

With a constant 53.095 kW auxiliary load, grid import is

$$P_t^{\text{grid}} = P_t^{\text{IT}} - P_{\text{dis},t}^{\text{UPS}} + P_{\text{ch},t}^{\text{UPS}} + P_t^{\text{ch,CRAC}} + P_t^{\text{ch,TES}} + P_{\text{aux}}. \quad (12)$$

The objective minimises signed energy settlement cost across the core and look-ahead,

$$\min \sum_t \pi_t P_t^{\text{grid}} \Delta t / 1000, \quad (13)$$

without storage degradation, standing losses, terminal-energy credit or network charges. The first annual interval starts with 600 kWh in the UPS, 500 kWh-th in TES and temperatures (28.5, 26, 20, 27)°C; no artificial daily cyclic condition is imposed.

The benchmark processes work on arrival and disables UPS and TES dispatch, while retaining cost-optimised cooling under the same thermal limits. The optimised case enables all three resources. Full-year one-at-a-time endpoints set flexible-workload share, UPS capacity and TES capacity to 0.5× and 1.5× their central values, preserving storage power ratings and initial energy fractions. Twelve representative weeks supply only the intermediate 0.75× and 1.25× response shape. Pyomo and HiGHS solve each MILP with a requested 0.1% relative gap, a 300 s daily limit and a normally accepted gap of 1%. The reported runs used Python 3.12.0, Pyomo 6.10.1 and HiGHS 1.15.1 on Windows 11 with an eight-core AMD Ryzen 7 9800X3D processor.

E. Representative-day flexibility assessment

A representative normal-length day is selected objectively. For every eligible day, price mean, dispersion and extremes; negative-price share; mean and peak grid import; CPU utilisation; cost; and opening UPS, TES and cold-aisle states are standardised against the annual distribution. The smallest root-mean-square standardised distance selects 22 September 2025.

The exact 96-interval annual trajectory and the following 12 intervals are loaded from the frozen annual result. For

a request starting at t_0 , the stored aggregate CPU path is fixed before the event and the six physical states at t_0 are pinned to their stored values. Deadline-level cohort allocation is reconstructed consistently with that aggregate path because the original annual export retained aggregate CPU rather than every intra-day cohort decision. Thus no aggregate workload or physical-state preconditioning is credited to the event.

For requested deviation ΔP , the event constraint is

$$\left| P_t^{\text{grid}} - (P_t^{\text{grid,base}} + \Delta P) \right| \leq 0.1 \text{ kW}. \quad (14)$$

Negative ΔP denotes upward flexibility (reduced import) and positive ΔP denotes downward flexibility (increased import). Start times are sampled every hour. Negative requests span -500 to -100 kW in 50 kW increments; positive requests are 25, 50 and 75 kW. The search builds a certified duration in one-hour increments, then refines the first failed interval to 15-minute resolution, up to 12 hours or the end of the local day. The first longer duration is also solved as a boundary check. For a feasible request, horizon-terminal UPS and TES energy must be no worse than the annual continuation and each terminal temperature must lie within 0.05°C of it. Equation (13) guides each trial towards a least-cost feasible dispatch. A 60 s screening limit is used per solve; a timeout without an incumbent is not classified as infeasibility. If the next 15-minute increment remains unresolved, the accepted duration is retained as a certified lower bound and marked “+” in the heat map. Selected component-decomposition cases permit an extended boundary check.

For interpretation, the realised grid deviation is decomposed additively as

$$\begin{aligned} \Delta P^{\text{grid}} = & \Delta P^{\text{IT}} + \Delta P^{\text{ch,CRAC}} + \Delta P^{\text{ch,TES}} \\ & + \Delta(P_{\text{ch}}^{\text{UPS}} - P_{\text{dis}}^{\text{UPS}}). \end{aligned} \quad (15)$$

Every event is audited for target tracking, recovery, workload conservation, objective reconciliation, mode exclusivity and component closure.

IV. RESULTS AND DISCUSSION

A. Central annual result and cost composition

The workload-at-arrival benchmark costs GBP 550,503.59 in 2025, whereas integrated optimisation costs GBP 522,707.23. The saving is GBP 27,796.37, or 5.049%. Table I reports all completed full-year endpoints.

The optimised case imports 6.794 GWh compared with 6.718 GWh in the benchmark. The saving therefore comes from moving load to cheaper signed-price intervals, not from reducing annual energy. Peak import rises from 974.9 to 1681.9 kW as storage charging and deferred-work recovery concentrate consumption. This is an energy-market result; capacity tariffs or a tighter grid constraint would change the dispatch.

Table II reconciles exactly to the settlement totals. Relative to the benchmark, time-shifted IT demand reduces signed cost by GBP 15,239 and CRAC operation by GBP 14,085. TES charging adds GBP 6,905, whereas the net UPS grid effect contributes a GBP 5,377 reduction. The fixed auxiliary contribution is unchanged. These are settlement contributions,

Table I
FULL-YEAR OPERATING-COST AND SENSITIVITY RESULTS.

Case	Cost (GBP)	Saving (%)	Change (pp)	Energy (GWh)
Baseline	550,504	0.000	–	6.718
Flexible workload 0.5×	531,117	3.522	-1.528	6.782
Flexible workload 1.0×	522,707	5.049	+0.000	6.794
Flexible workload 1.5×	516,058	6.257	+1.208	6.807
UPS capacity 0.5×	525,374	4.565	-0.484	6.771
UPS capacity 1.5×	520,222	5.501	+0.452	6.815
TES capacity 0.5×	524,080	4.800	-0.249	6.784
TES capacity 1.5×	522,057	5.167	+0.118	6.799

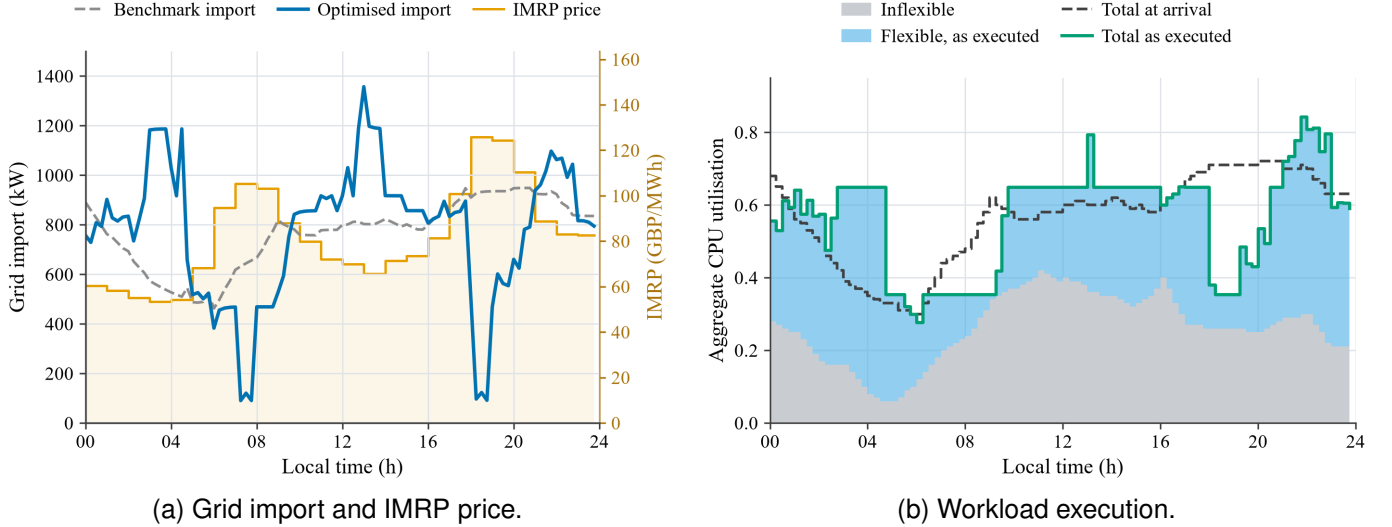


Fig. 3. Annual benchmark and optimised operation on the objectively selected representative day, 22 September 2025. In (b) the shaded bands separate inflexible from flexible computation as executed, and the dashed line is the same day's workload at arrival.

Table II
ANNUAL SIGNED SETTLEMENT-COST COMPOSITION.

Component	Baseline	Optimised (GBP)	Change
IT electrical demand	407,653	392,414	-15,239
CRAC chiller	105,337	91,251	-14,085
TES charging chiller	0	6,905	6,905
UPS net grid effect	0	-5,377	-5,377
Auxiliary overhead	37,514	37,514	0
Total settlement cost	550,504	522,707	-27,796

not stand-alone asset profits: each value depends on the jointly optimised dispatch.

Figures 3 and 4 are slices through the linked annual trajectory. Their storage, thermal and workload opening conditions are inherited from 21 September rather than reset.

B. Sensitivity and asset ranking

Flexible-workload share is the dominant tested parameter. Halving it reduces the annual saving to 3.522%, while increasing it by half raises the saving to 6.257%. UPS capacity ranks second, spanning 4.565–5.501%, and TES capacity third, spanning 4.800–5.167%. The frozen ranking is therefore flexible workload, UPS energy capacity and TES energy capacity.

The sampled central saving is 6.713%, compared with 5.049% annually. Figure 5 therefore centres the sampled

curves on their own central case; it does not present sampled percentages as annual estimates or infer interactions from a one-at-a-time design.

C. Recovery-certified flexibility envelope

Figure 6 shows a strongly time-dependent and asymmetric envelope. The longest response is a +25 kW import increase beginning at 06:00, sustained for 9.75 hours; the +50 and +75 kW requests at the same time last 8.00 and 7.75 hours. Reduced-import service reaches 5.25 hours at 13:00 for -100 and -150 kW requests, and also at 02:00 for -100 kW. Larger magnitude does not imply a monotonic duration at every start because the request is an exact signed grid target in a mixed-integer system: it changes which combinations of delayed work, discrete storage modes and thermal recovery are reachable.

Figures 7 and 8 show that the same external request can be delivered by different internal portfolios. Reduced import combines IT deferral, lower CRAC/TES charging demand and UPS discharge, followed by compensating changes within the event and recovery horizon. Increased import is obtained mainly by advancing feasible work and raising its associated CRAC demand, with TES charging contributing when thermal headroom is available. The component bars close to the requested grid deviation to within 4.4×10^{-11} kW; the

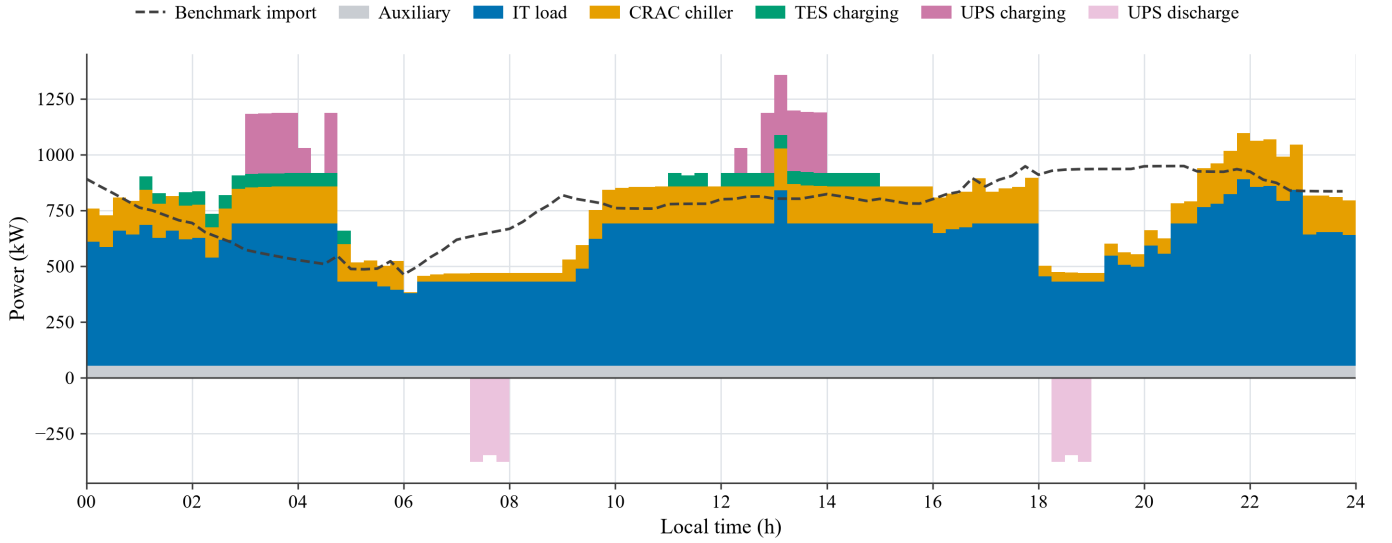


Fig. 4. Optimised component dispatch on 22 September 2025. The stack is an exact decomposition of optimised grid import: bars above the axis are metered demand and the band below the axis is UPS discharge, which supplies IT behind the meter. The dashed line is benchmark grid import.

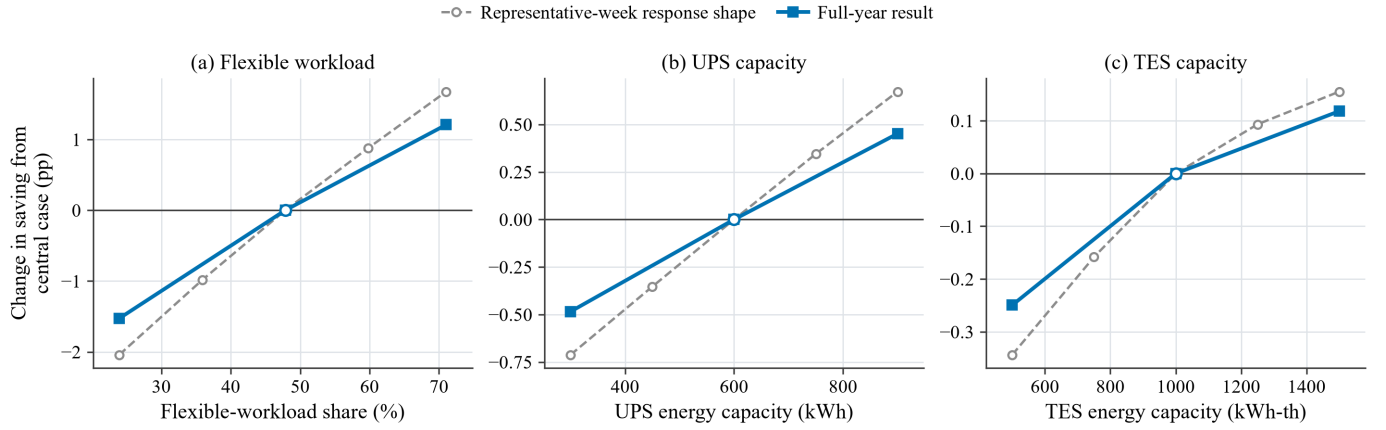


Fig. 5. Sensitivity response relative to each central case. Solid blue endpoints are full-year runs; dashed grey intermediate curves are from twelve representative weeks and describe response shape only. The open marker in each panel is the shared central case.

largest event tracking error is 0.1 kW and the largest terminal-temperature error is 0.05°C.

D. Continuity, solver quality and terminal treatment

Every annual chain passes timestamp coverage, state hand-off, workload conservation, charge/discharge exclusivity and settlement reconciliation. The few time-limited daily solves retain feasible incumbents. The largest exceptional relative gap occurs in the low-flexibility case on 25 May: a 900 s rerun leaves a 5.652% relative gap because its negative objective is close to zero, but the absolute incumbent-bound width is only GBP 0.102. Its committed cost changes by about GBP 0.002, its downstream state is unchanged and all physical audits pass; it is retained with this qualification.

At midnight after 31 December, 0.642875 CPU-hours remain within deadline and are completed in the actual 1 January 2026 look-ahead. Calendar-year reporting includes only 2025 core intervals; look-ahead cost is a feasibility certificate. A continuation-value test spanning the median to the maximum observed 1 January price changes annual cost by at most

0.000381%. This supports the zero-value central convention while exposing its consequence.

The 288-cell flexibility grid uses a uniform hourly start-time mesh. All event dispatches meet the accepted gap and all physical audits pass. Exact duration boundaries are established for 283 cells. Five cells retain certified lower bounds because the next 15-minute increment did not resolve within the 60 s screening limit; these are marked explicitly rather than treating a timeout as infeasibility.

E. Scope and limitations

This is a stylised operating experiment, not a forecast for a particular facility. Ambient temperature, COP and the repeated local-clock workload shape are fixed; storage ageing, standing losses, network tariffs and service revenue are omitted. The exact-target event formulation is intentionally conservative and does not model a market tolerance band beyond 0.1 kW. The representative day demonstrates state-consistent duration capability but is not an annual availability distribution. Finally,

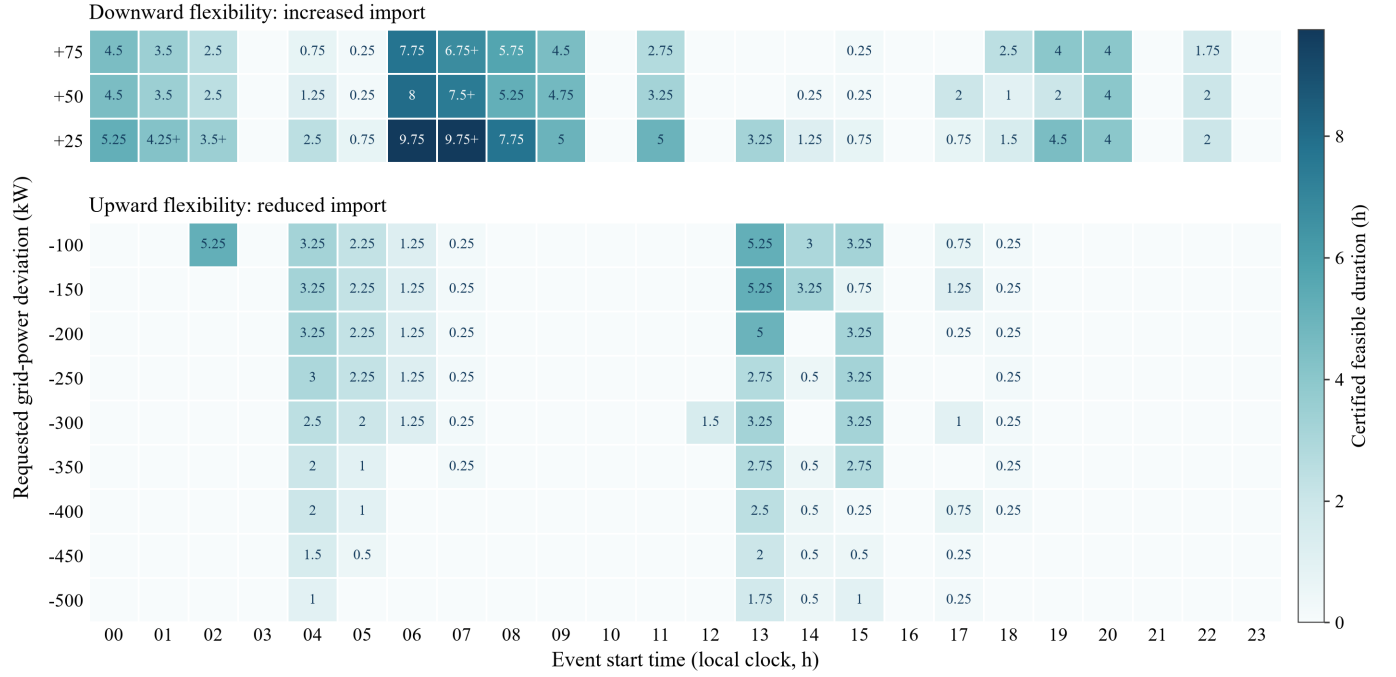


Fig. 6. Recovery-certified duration on 22 September 2025 for every start time and requested deviation. Blank cells carry a certified duration of zero. A “+” marks a certified lower bound where the next 15-minute increment was unresolved at the screening limit.

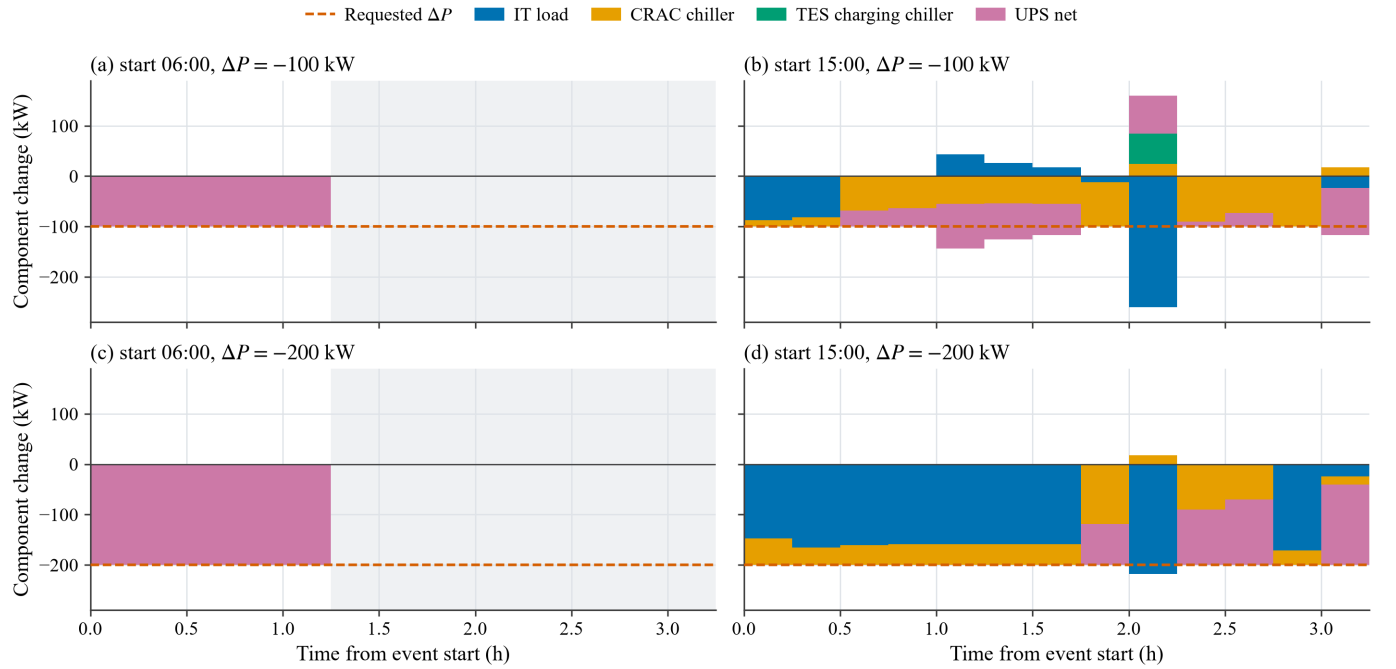


Fig. 7. Component decomposition for selected upward-flexibility events (reduced import). Bars are stacked above and below the axis according to the sign of each component change and sum to the dashed requested deviation. All panels share one time axis; the grey band marks time beyond the certified duration of that event.

component contributions are dispatch dependent and should not be interpreted as independently additive asset values.

V. CONCLUSION

Linked rolling optimisation across 2025 reduces modelled electricity cost from GBP 550,503.59 to GBP 522,707.23, a saving of 5.049%, while preserving physical and workload

continuity through every local day and daylight-saving transition. Full-year endpoints identify flexible workload as the most influential tested resource, followed by UPS and TES energy capacity. The additive annual cost table shows how IT rescheduling and CRAC timing produce most of the gross reduction, while TES and UPS contributions reflect their joint dispatch.

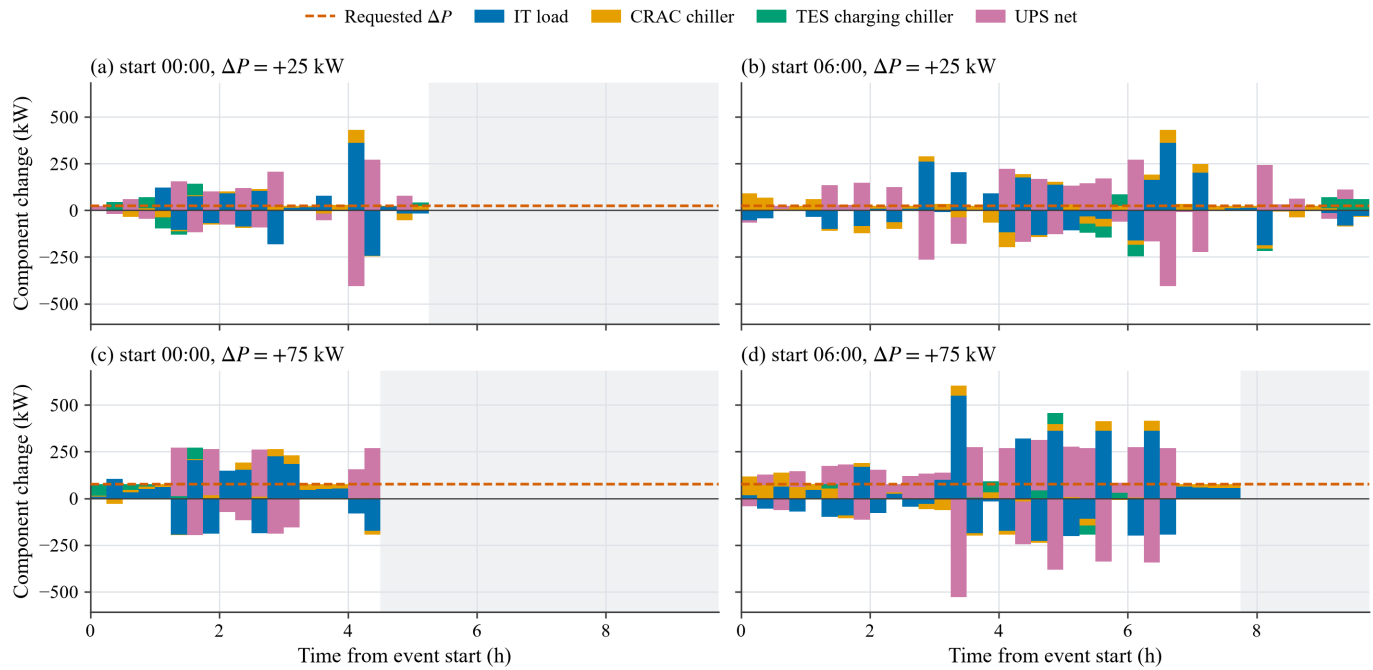


Fig. 8. Component decomposition for selected downward-flexibility events (increased import). Notation is as in Fig. 7.

Starting from the same frozen annual trajectory, the representative-day study converts this economic flexibility into a recovery-certified grid-service envelope. Increased import reaches 9.75 hours at +25 kW, while reduced import reaches 5.25 hours at -100 and -150 kW. Component decompositions demonstrate that equivalent grid responses can use materially different mixes of workload, cooling, TES and UPS flexibility. This dependence on start time, direction and internal state is precisely why independent reset-day estimates are insufficient for credible flexibility offers.

Future work should extend the linked formulation to weather-varying cooling performance, battery degradation, network constraints, service revenues, multi-day availability statistics and factorial interactions among resources.

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